

Adaptive Trial Design for the RAS-Inhibitor Era: Combinations, Resistance, and Biomarker-Guided Therapy

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Today's argument

New RAS therapies in PDAC need new kinds of trials.

Methods to optimize combination doses are established. Designs that adapt to emerging resistance are enrolling.

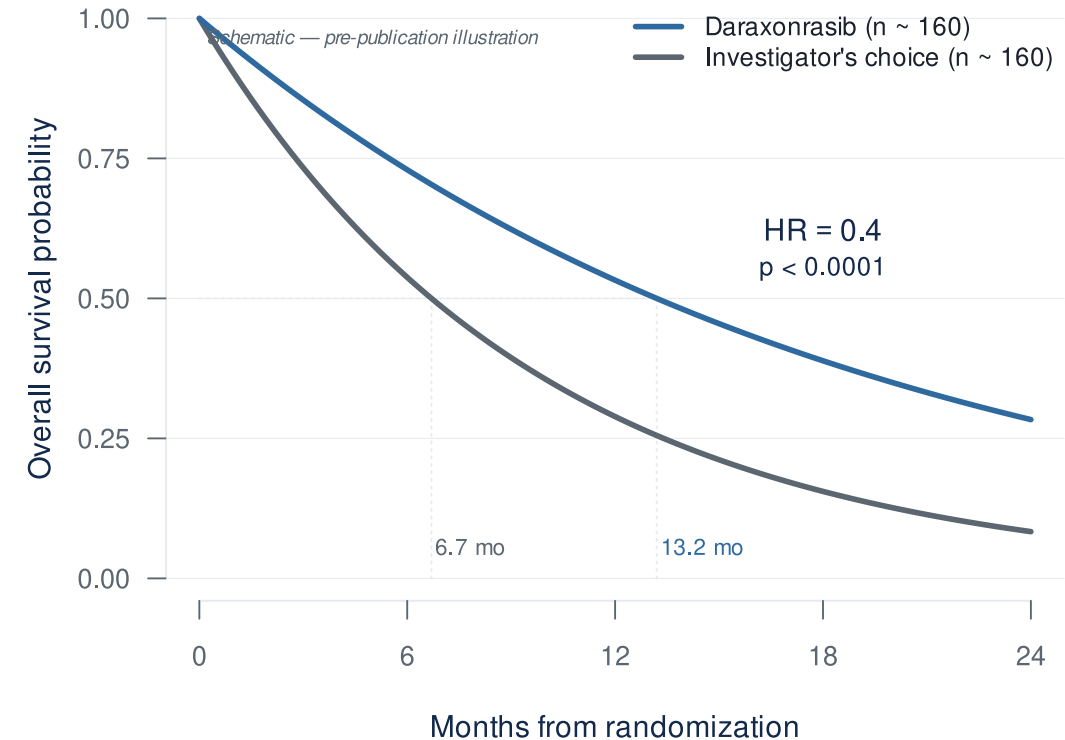
The difficulty: getting them calibrated and launched.

The clinical case

Why combinations, and why now?

The RAS revolution has arrived for PDAC

- RASolute 302 (2L mPDAC, *NEJM* 2026)¹:
 - **Median OS: 13.2 vs 6.7 months**
 - **HR 0.40 — 60% reduction in risk of death ($p < 0.0001$)**
- Daraxonrasib (RMC-6236) — oral, multi-selective RAS(ON) inhibitor
- ASCO 2026 Plenary, May 31 (Wolpin)



...but monotherapy has visible ceilings

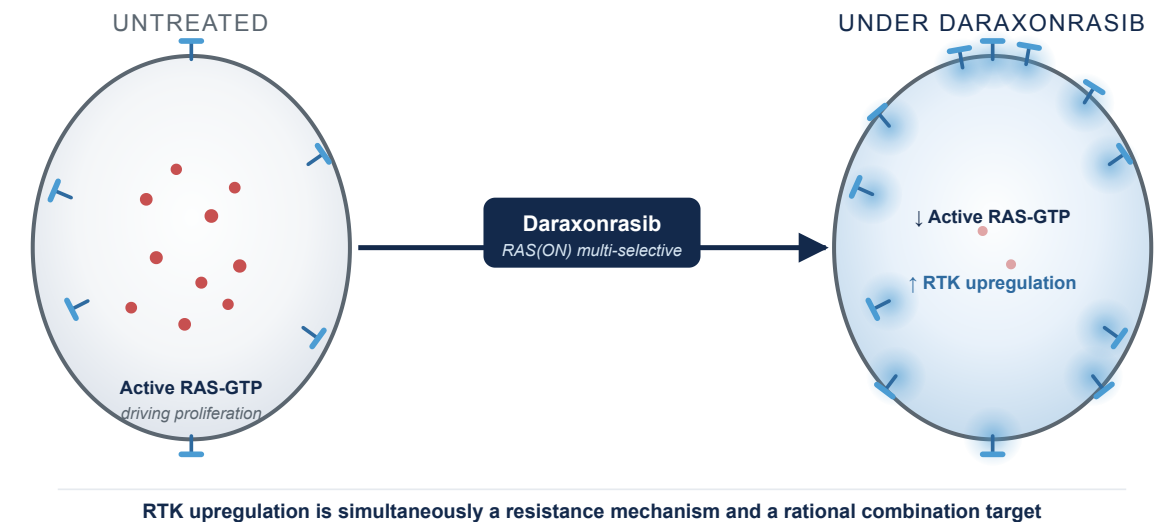
RMC-6236-001 (PDAC cohort, NEJM 2026)²



Half of patients on monotherapy at 300 mg already need dose adjustments. Combinations layer on top — with less room to spare.

Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface — the mechanistic case for biomarker-driven, resistance-preemptive combinations³
- **1L combination cohort** (daraxonrasib 200 mg + GnP, n=40)⁴:
 - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**



The dose-finding problem

Why MTD only may not be best for RAS combinations, and what replaces it.

Why combinations break the rules you grew up on

- **Rivière's review of 162 combination Phase I trials: 88% used 3+3** even when both drugs were escalated^{5,6}.
- **Why 3+3 doesn't fit combinations.** Built for single-agent cytotoxics⁷, it assumes monotonic dose-toxicity, Cycle-1-observable DLTs, and a single dose dimension — combinations strain all three.
- **The two-drug dose map is never built.** In practice: pick A's dose, fix it, add B. Two monotherapy escalations run in series; the joint efficacy/toxicity behavior across (A, B) pairs (the “joint surface”) is never characterized.

Two-drug dose map: how efficacy and toxicity change across every (A, B) dose pair — not just along one drug's axis.

The Amgen case: sotorasib + pembrolizumab

Same sponsor, two different designs:

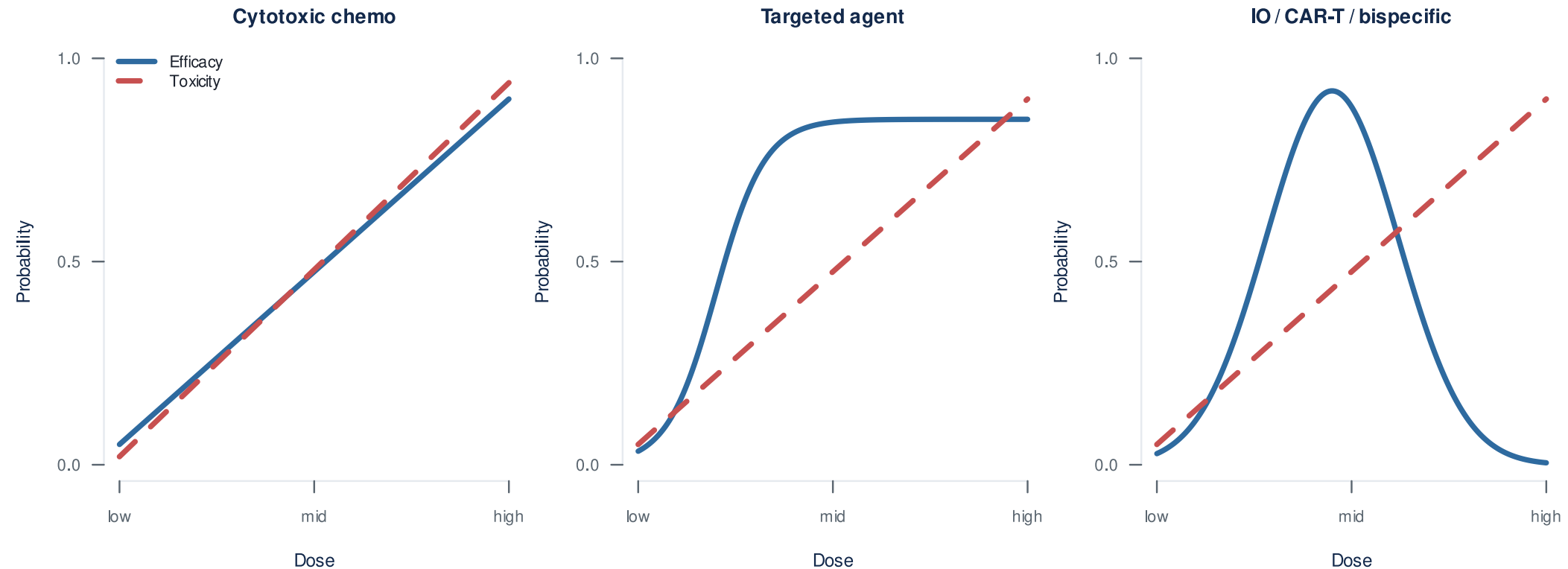
- **Monotherapy (CodeBreak 100):** Bayesian model-based, single-agent surface
- **Combination:** IO dose fixed, only sotorasib varied — joint surface never modeled

The defining toxicity:

- Severe **immune-mediated** hepatotoxicity, not predictable from the single agent
- **88%** of Grade 3-4 events outside the Cycle 1 DLT window⁸

What MTD-only dose-finding was never built to handle, all at once: joint surface unsearched, non-additive toxicity, late-onset toxicity.

Why MTD-only dose-finding doesn't fit targeted agents and immunotherapies



The cost of MTD-only dose-finding

- **Sotorasib** (KRAS G12C, first-in-class, approved 2021)¹⁰:
 - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
 - FDA required a **240 mg vs 960 mg** comparison. **PFS was similar** at one-quarter the exposure¹¹
- **MTD $\neq$ optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions¹²

Same drug class as daraxonrasib.

Project Optimus: the regulatory ground has shifted

How OBD selection actually works

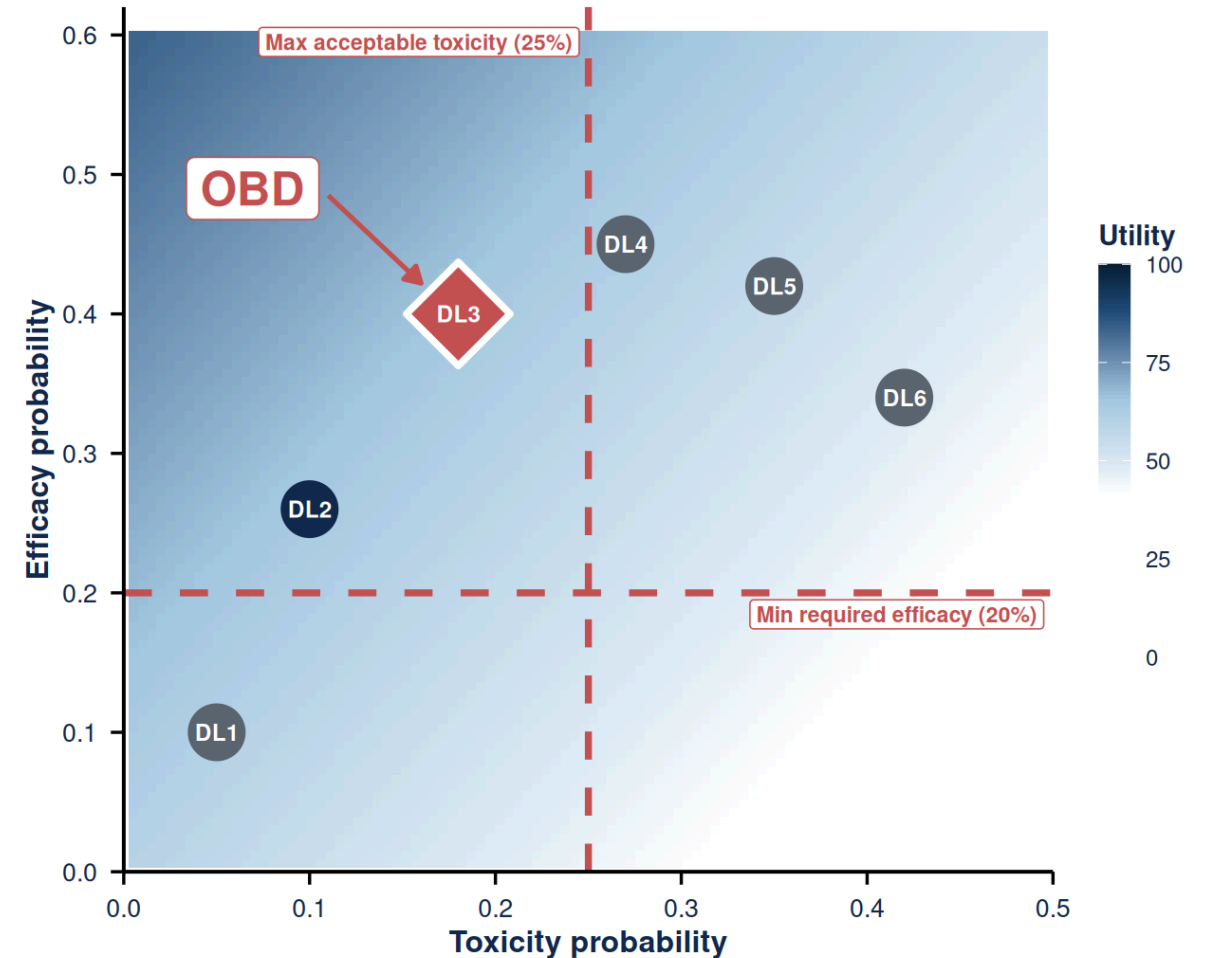
Instead of escalating until toxicity, choose the dose with the best benefit-risk profile.

“A response is worth this; a Grade 3 toxicity costs this.”

The team writes that utility table down up front. The design then searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

Elicited utility table (Zhou 2019 framework)¹⁴:

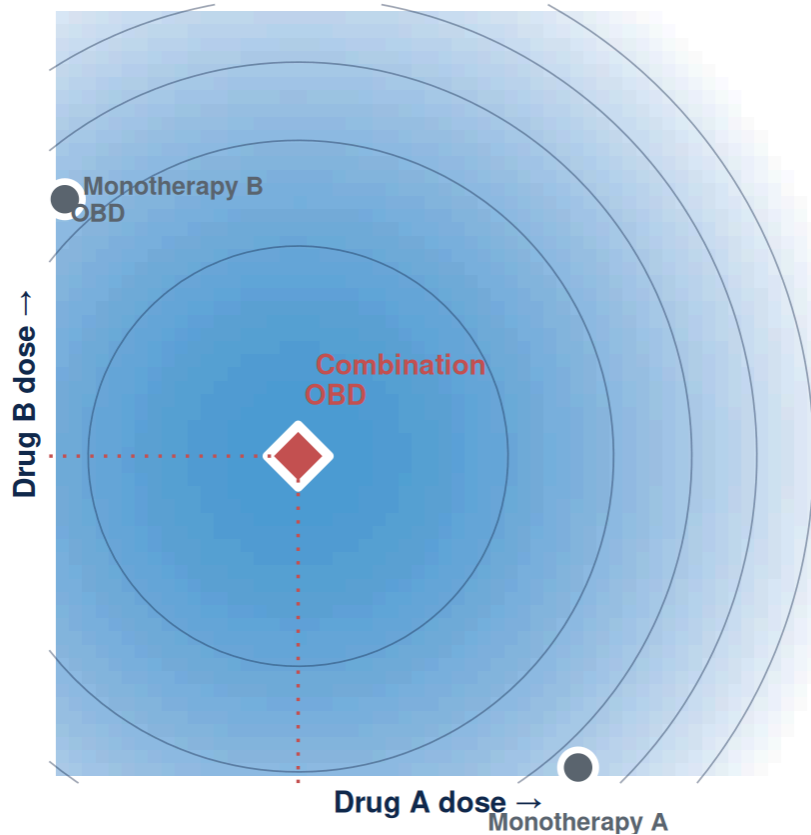
	TOX	NO TOX
Response	40	100
No response	0	60



OBD = admissible dose (inside red lines) with the highest utility.

Combination dose selection in RAS is harder

Combinations make MTD-only dose-finding worse on every axis.



- The combination OBD can sit **below both single-agent OBDs**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **toxicity margin is already spent** at the monotherapy dose

Fix what you know. Search what you don't.

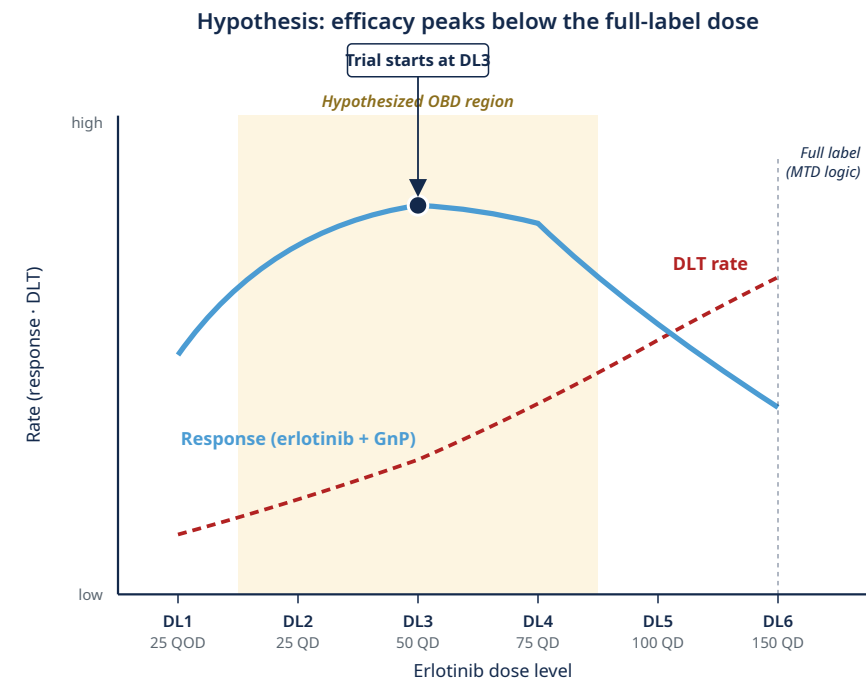
RMC-GI-102 (daraxonrasib + GnP) fixed daraxonrasib at 200 mg without searching the joint surface⁴.

Conceptual schematic of the benefit-risk surface. The combination OBD sits below both single-agent OBDs; single-agent data alone cannot reconstruct the contour.

Finding the right erlotinib dose in basal-like PDAC

- **Clinical question.** Could *lower*-dose erlotinib work *better* than full-dose erlotinib in basal-like PDAC, when combined with GnP?
- **Design.** First find the safe dose range, then choose the dose with the best response/toxicity trade-off^{14,17}.
- **Why it matters.** This is an OBD-below-MTD question; 3+3 cannot answer it.

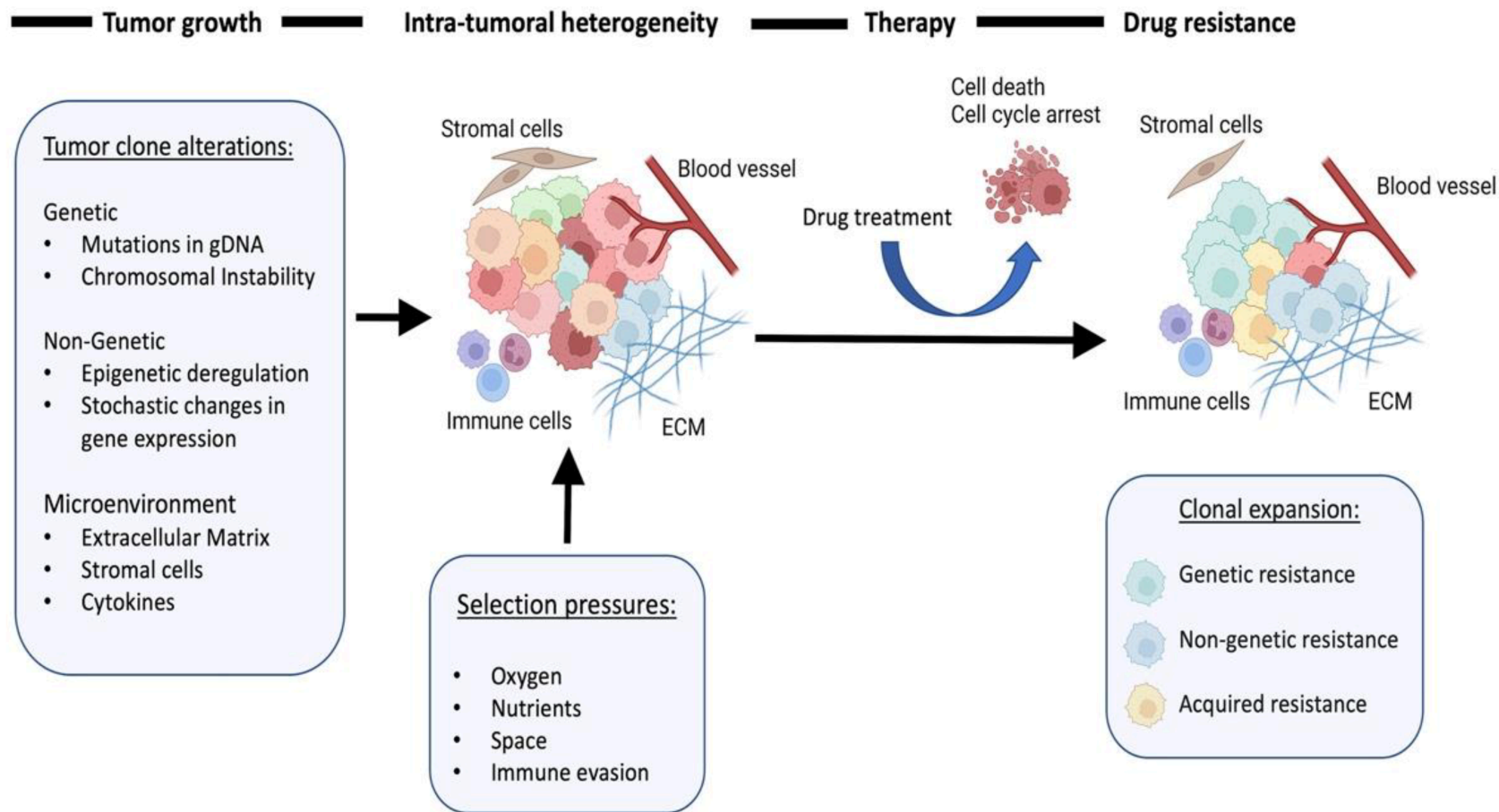
Trial: LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by PurlIST¹⁸ . combination erlotinib + biweekly GnP · OBD-below-MTD hypothesized from prior data¹⁹ · **up to 52 patients; ~28 expected at the OBD.**



Resistance is the next problem

Phase 2 onward: each patient is a moving target

Resistance is a dynamic process

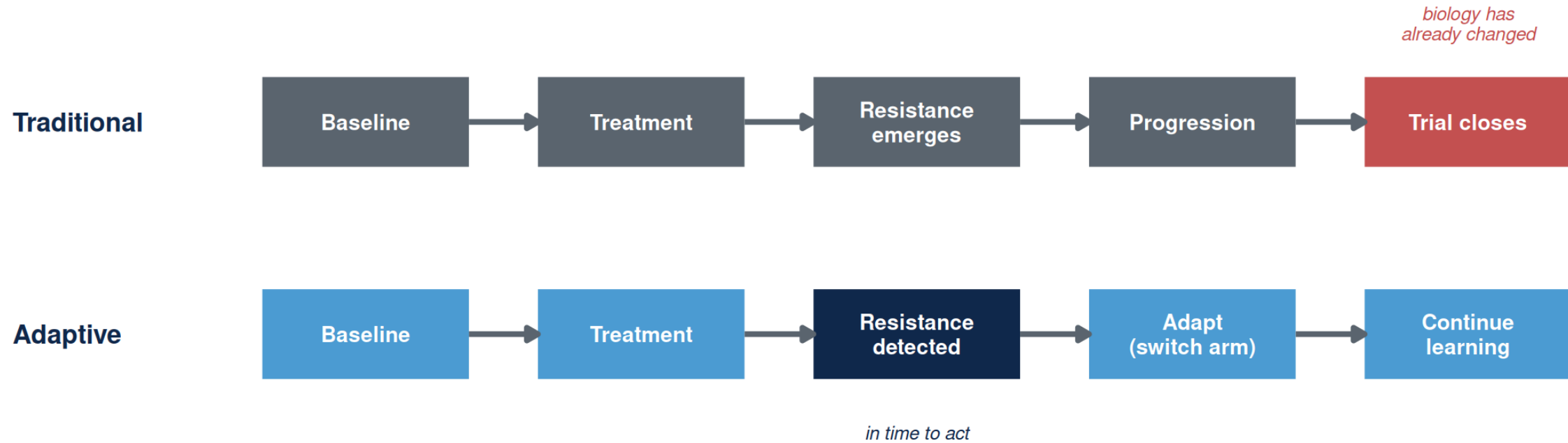


Resistance mechanisms emerge under RAS-inhibitor pressure

Daraxonrasib monotherapy in mPDAC (n=44, paired ctDNA): 59% show RAS-pathway resistance at progression — KRAS amplification in 36%²¹

Adapted from Karagiannis & Rampias, Cancers 2022²⁰.

Why classical trial designs miss resistance



RESISTANCE TYPE	WHAT YOU NEED
Genomic (KRAS amplification, NRAS/BRAF/MAP2K1)	ctDNA
Cell-state / plasticity (basal↔classical, partial EMT)	biopsy + transcriptomics + imaging

A resistance-aware trial has to **do both** during enrollment^{22,23}.

ADAPT / EVOLVE: what a resistance-aware trial looks like

ARPA-H's resistance-tracking template.

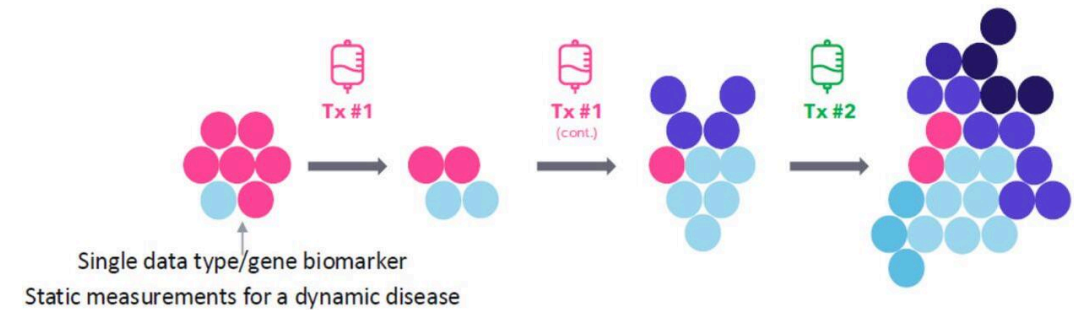
- Longitudinal resistance detection
- Biomarker-guided adaptation
- Adaptive arm-swaps on emerging biology

ARPA-H ADAPT → EVOLVE (UNC-led; Carey MPI, Rashid MPI).
Enrolling.

ADAPT Goal: Identify and target changes in metastatic cancers

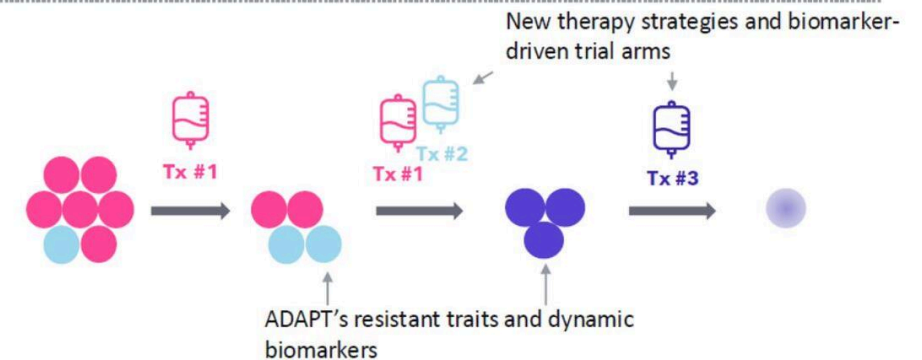
Current State

- Cancer changes during treatment. Therapies **not** often informed by tumor change to cancer resistance traits.
- There are only **limited** tumor measurement biomarker analyses, which are most often performed **prior to** treatment.



ADAPT's upcoming "future state"

- Therapies **will be** informed by and tailored to observed cancer resistance traits using biomarkers.
- **In-depth** tumor measurements will be taken **before, during and after** treatment.



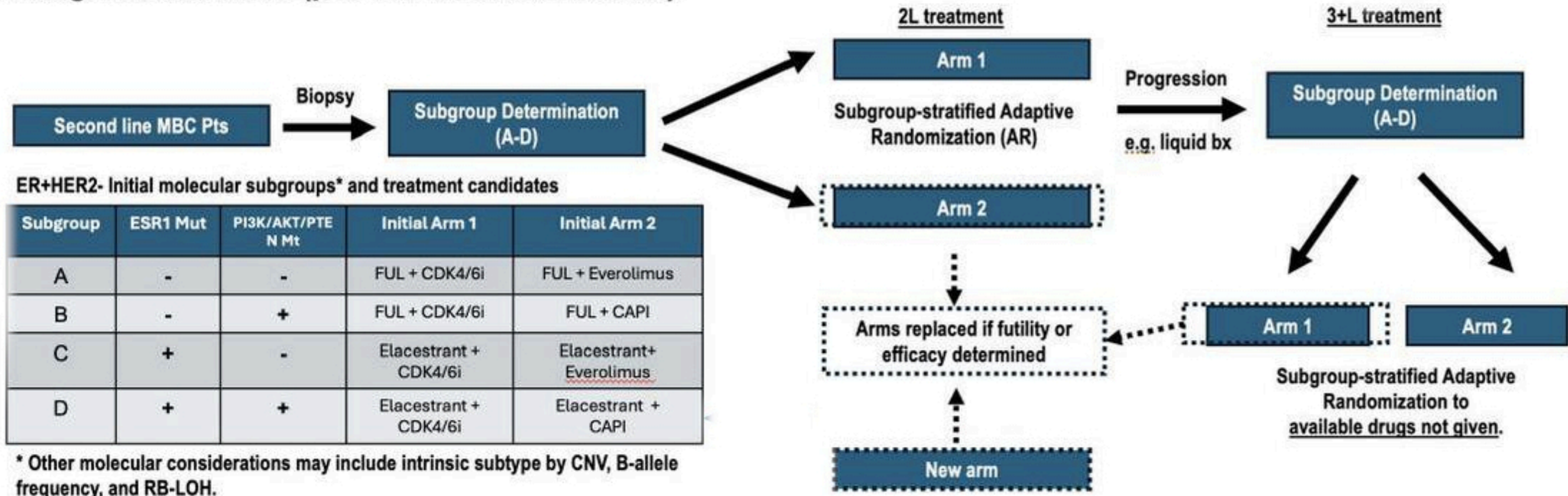
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ARPA-H ADAPT — current state (top) vs dynamic-biomarker state (bottom).

EVOLVE-BDT — and what this would look like in PDAC

EVOLVE is the operational template; PDAC is the next application.

First generation trials (pre-TA1 novel biomarkers)



TBCRC EVOLVE-BDT flow — biomarker-stratified subtrials, adaptive randomization, in-trial arm replacement on futility or efficacy.

EVOLVE-BDT (real, enrolling)

ER+/HER2– and TNBC, biomarker-stratified subtrials with Bayesian arm-swap on futility/efficacy.

PDAC analog (hypothetical)

Basal-like vs classical by PurlST¹⁸, same Bayesian arm-swap logic on daraxonrasib-combination arms.

Same architecture, different disease.

What this demands of the statistical design

Four design axes any such trial needs.

Adaptive stopping

Bayesian futility /
efficacy boundaries
at each interim

Real-time biomarker discovery

Pre-specified signatures
+ agnostic
discovery window

Longitudinal monitoring

Serial ctDNA,
imaging, biopsies
on every patient

In-trial evaluation

Biomarker-guided
cohorts within the
same protocol

Methods are ready. We're operationalizing them in PDAC trials — starting with PANGAEA.

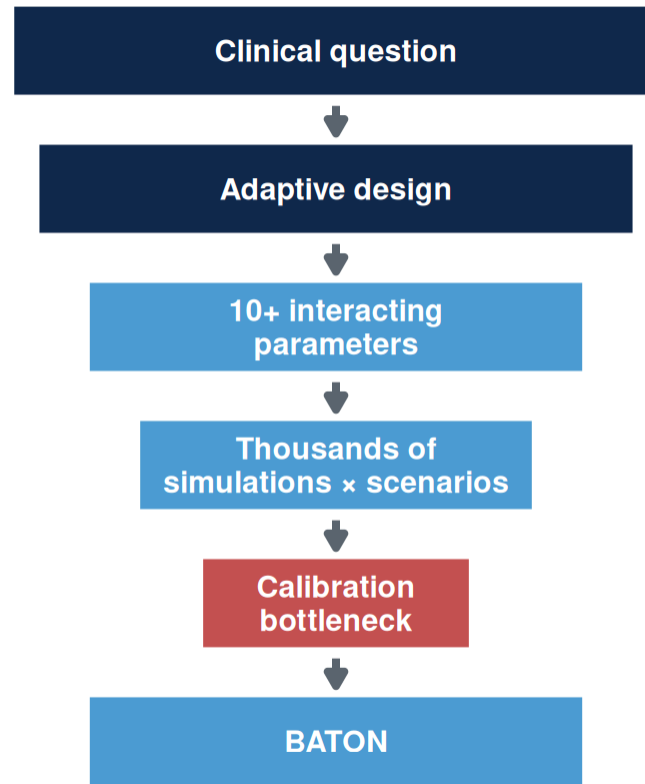
Putting these into practice

Why don't we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.

*Calibration tunes a design to hit its **operating characteristics** — the trial's report card: false-positive control, power, expected enrollment, early stopping, and time to readout.*



Bayesian designs adopted 75%

Dose-optimization plans 30%

Methods adopted. Dose-optimization plans — where calibration cost bites — not²⁴.

Our group: several months per design. For combined dose + resistance designs, hand-tuning is infeasible.

The calibration challenge: four competing goals

Designing an adaptive trial requires achieving all four simultaneously.

1. FDA compliance

Power ≥ 0.80
Type I ≤ 0.10

2. Minimize patients

Smaller $E[N]$
Efficient use of
scarce patients

3. Stop bad early

Futility stopping
Protect patients
from inactive arms

4. Accelerate good

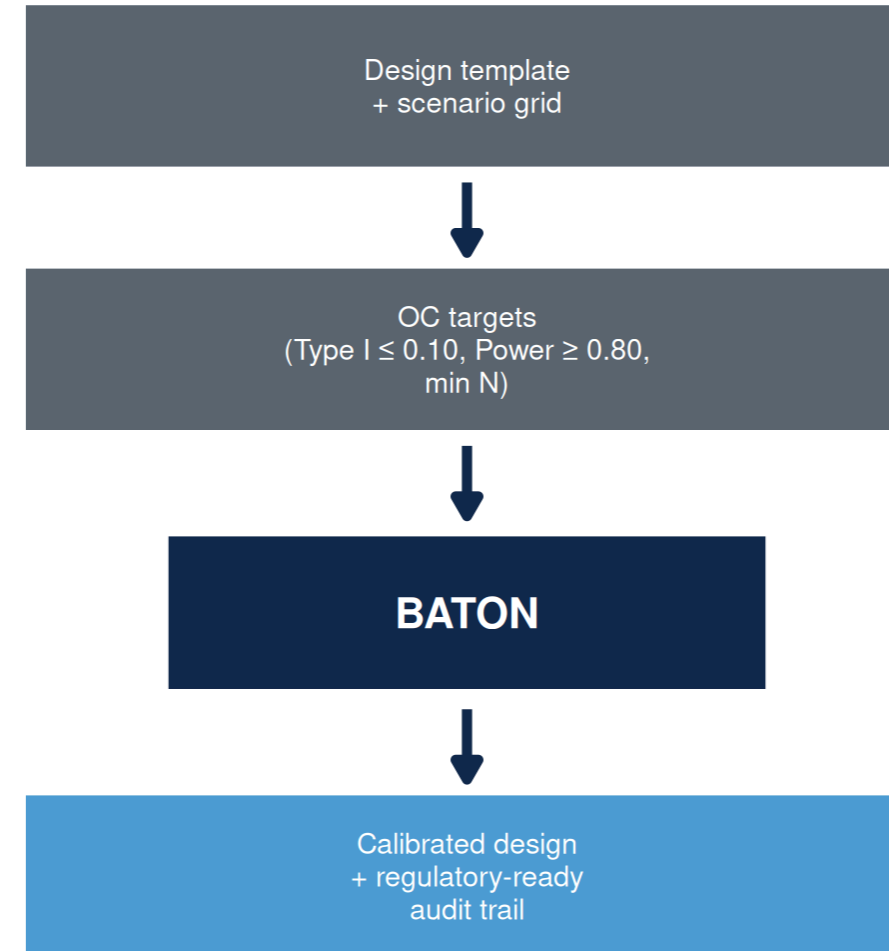
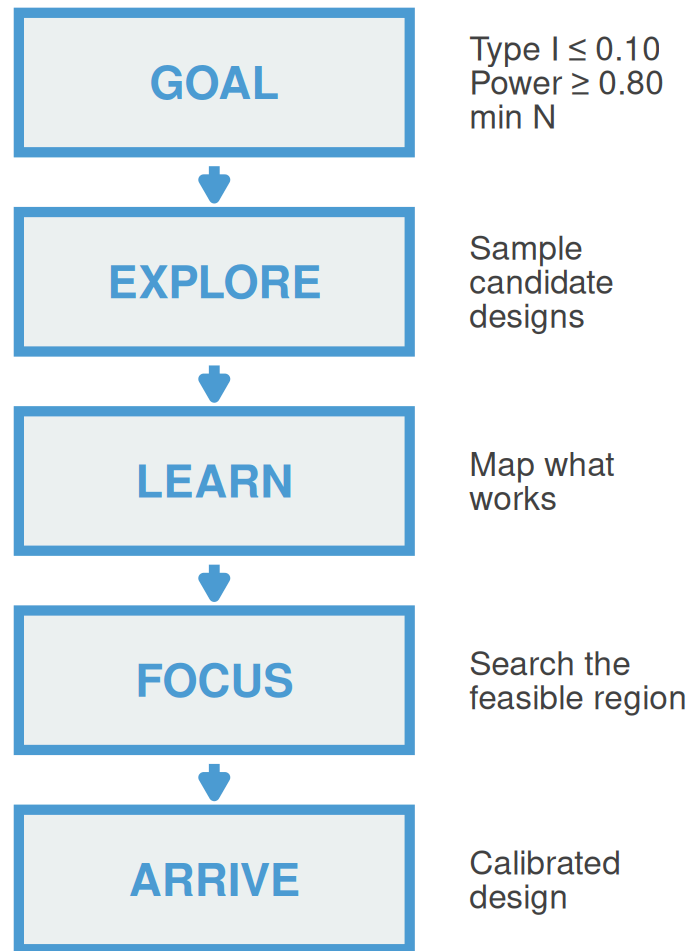
Efficacy stopping
Faster access
when drug works

***“We could design the science.
We needed a faster way to
calibrate the trial.”***

BATON: the calibration tool

BATON = Bayesian Adaptive Trial Optimization²⁵

BATON turns clinically necessary trial complexity into calibrated, launchable designs.



You bring the clinical inputs; BATON returns the calibration.

You bring the clinical design; BATON evaluates hundreds of candidates by simulation and returns the ones that meet your targets. R package: github.com/naimurashid/BATON²⁵

A BATON-calibrated design hits all four goals — for PANGEA Phase II

1. FDA compliance ✓

Type I 7.0% (target $\leq 10\%$)
Power 82.2% (target $\geq 80\%$)

2. Minimize patients ✓

~23 patients spared
(avg N 53 vs Max N 76)

3. Stop bad early ✓

78% early futility stop
when drug is inactive

4. Accelerate good ✓

81% early efficacy stop
when drug works

Far fewer evaluations than grid search — finds feasible designs that grid search often misses entirely. For a 4-parameter design like this one: manual calibration takes days, BATON takes minutes, with a regulatory-ready audit trail.

What this means for a PDAC patient

Mr. Hernandez, 62, mPDAC, basal-like by PurlST, progressed on 1L GnP.

Traditional fixed Phase II

- All **76 patients** enrolled before any look
- Drug effect discovered only at study end
- **~36+ months** to read out

PANGEA Phase II (BATON-calibrated)

- **78% chance** of early futility stop **when the drug is inactive**
- Average patients enrolled: **~53** if inactive / **~58** if active
- **27.3 months** expected duration

Sooner answer, fewer patients on a losing arm — without losing power.

PANGEA Phase II: calibrated by BATON

BATON identified a design that preserves power, controls false positives, and allows early stopping when erlotinib isn't helping.

- After Phase I OBD → **Phase II Bayesian adaptive RCT**: GnP + erlotinib (at OBD) vs GnP alone, primary PFS
- **Design assumes** reference 5.5 mo → 9.2 mo (HR 0.58)²⁶
- **BATON-calibrated** four parameters under Type I ≤ 0.10 , Power ≥ 0.80 ²⁵

Achieved operating characteristics:

METRIC	VALUE
Power	82.2%
Type I error	7.0%
Max N	76 (82 accrual w/ buffer)
Expected N under H_0	53.0
Expected N under H_1	58.1
Pr(early futility stop H_0)	77.9%
Pr(early efficacy stop H_1)	81.4%
Expected duration	27.3 mo

H_0 = drug inactive (no PFS benefit over GnP alone) · H_1 = drug active at the assumed effect size (HR 0.58)

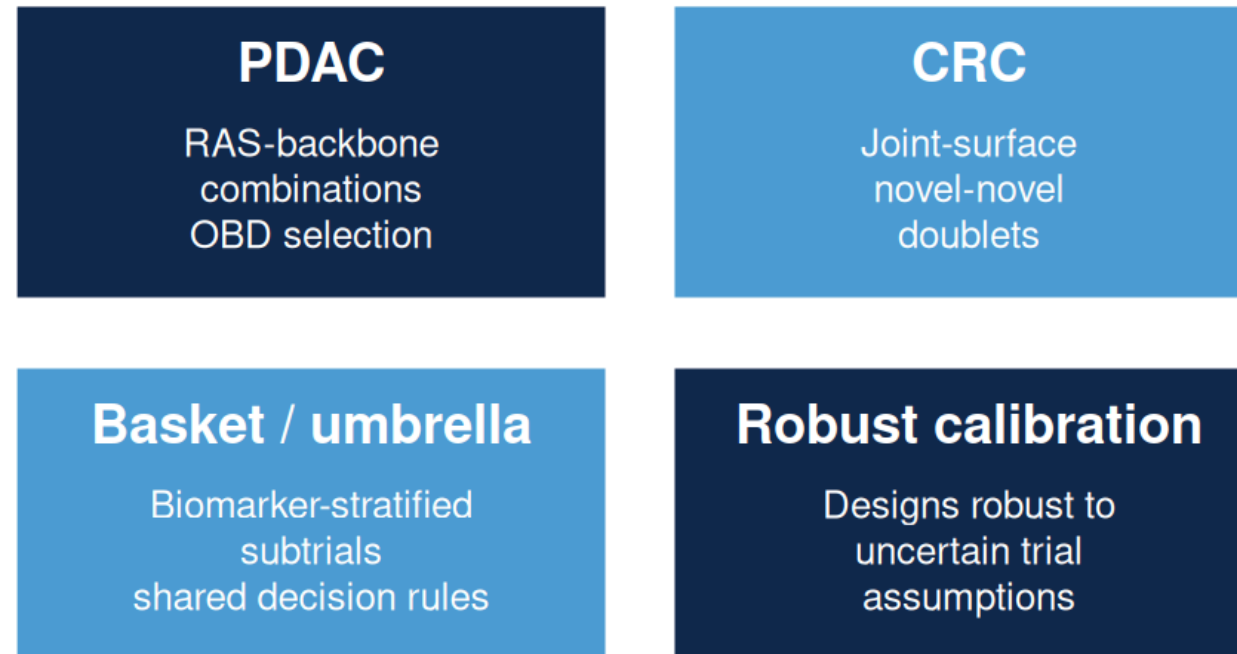
Read as: 78% chance of stopping early when the drug is inactive; 81% chance of stopping early when it works.

Without BATON: four interacting parameters rarely hit both constraints simultaneously. **With BATON:** joint calibration against both constraints, ~200 evaluations.

***An unexpected finding:** caps set too low suppress early-efficacy declarations — even when the drug works, the trial waits for the final analysis to say so. BATON returned the **best trade-off curve** across candidate caps; we chose 76 to preserve early-efficacy stopping. Manual calibration would have anchored on the first feasible design.*

Where this is going

BATON calibrates **aggressive**, **conservative**, or **balanced** designs — and ports across GI applications:



BATON is the infrastructure that makes adaptive trial design routine across the GI portfolio.

Take-home

1. **The RAS era needs new kinds of trials** — biomarker-guided, dose-optimized, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs can handle dose optimization, biomarker guidance, and resistance-aware adaptation.
3. **BATON makes them practical to calibrate and launch.**

Acknowledgments

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- Ashwin Somasundaram (PANGAEA PI)

EVOLVE MPI team

- Charles Perou (UNC) · Ian Krop, Eric Winer (Yale) · Antonio Wolff (Hopkins)

Methods collaborators

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Questions